Predicting Adverse Drug Reactions Using Deep Learning Methods

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Abstract

Adverse drug reactions (ADRs) are a leading cause of preventable patient harm worldwide. This report describes a machine learning pipeline for predicting patient-level ADR risk using the MIMIC-IV clinical dataset, restricted to the top 20 most commonly administered drugs. We evaluate and compare four classical machine learning baselines against three deep learning architectures, a Multilayer Perceptron (MLP), a residual network (ResNet) and an attention-based model, identifying the most promising approach using threshold-independent metrics and analyzing performance at a clinically motivated decision threshold. Our methodology is designed to be clinically interpretable and robust to the severe class imbalance inherent in ADR datasets (≈13.9% positive rate).

1 Introduction

This project addresses the challenge of predicting adverse drug reactions (ADRs) at the individual patient level. The end-to-end pipeline encompasses data extraction and labeling from MIMIC-IV, feature engineering, training of multiple classical and deep learning classifiers, and rigorous evaluation using both threshold-independent and threshold-dependent metrics.

Because ADR incidence in real clinical populations is rare relative to non-events, both modeling and evaluation require careful attention to class imbalance. We compare a suite of models spanning logistic regression, gradient boosted trees, and three deep learning architectures to identify which approach best captures the complex, heterogeneous signals present in electronic health records.

Figure 1 summarizes the end-to-end workflow.

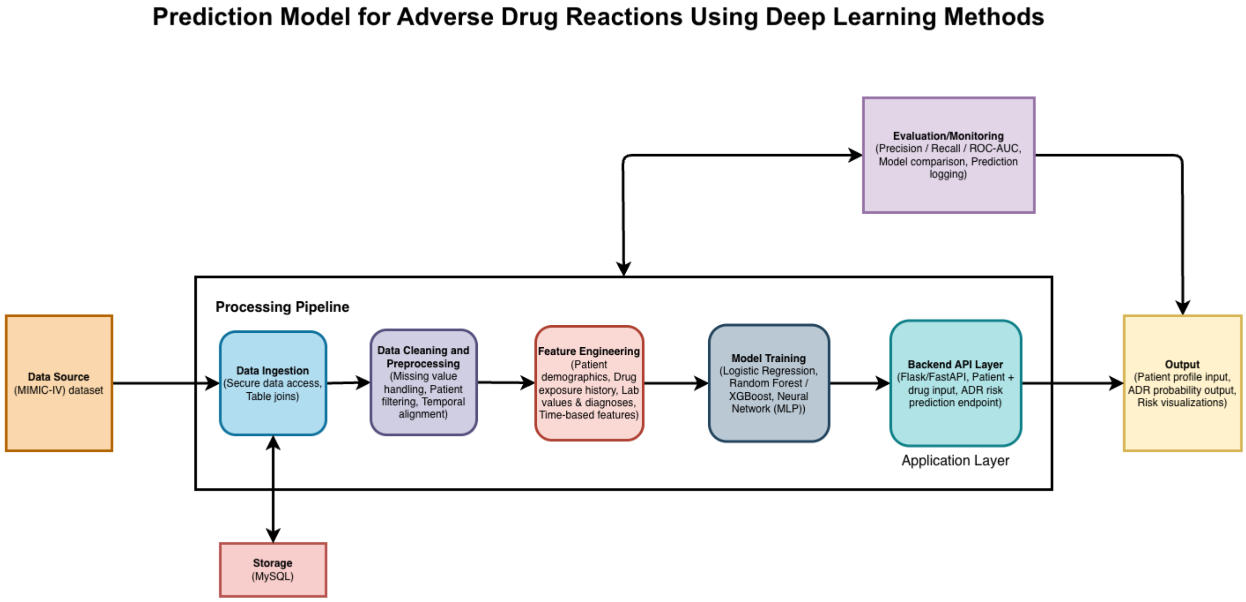


Figure 1: End-to-end project pipeline for ADR prediction.

2 Purpose of the Methodology

The methodology of this project is designed to effectively address the challenge of predicting adverse drug reactions at the individual patient level. We employ both classical machine learning and deep learning approaches in order to empirically compare results across a wide variety of model families.

The methodology contributes to solving the problem in several key ways. First, it handles the complexity and heterogeneity of clinical data in MIMIC-IV through systematic data cleaning, ADR labeling, and domain-informed feature engineering. Second, it restricts the drug space to the top 20 most commonly administered drugs in the dataset, ensuring that each drug class has sufficient event counts to support meaningful learning. Third, it sweeps a broad range of model families to identify the strongest approach empirically. Fourth, it delves more deeply into the most promising deep learning model through ablation and comparison studies that examine how results change when including different subsets of the dataset. Fifth, we tune decision thresholds on the validation set rather than defaulting to 0.5 to produce operating-point metrics appropriate for the imbalanced clinical setting. Throughout, we consider the specific needs of the healthcare industry, particularly the asymmetric cost of false positives versus false negatives.

Comparing multiple models is important because the complexity and heterogeneity of real-world clinical data make it non-obvious which model family will perform best. Classical models are interpretable and computationally efficient, but may fail to capture non-linear feature interactions. Deep learning models are more expressive but require careful regularization. Ultimately, this methodology enables a systematic and evidence-based approach to selecting models that can deliver accurate, patient-specific ADR risk predictions while remaining mindful of the operational constraints of clinical deployment.

3 Problem Statement

3.1 Defining the Problem

Because adverse drug reactions are recorded in the dataset as categorical variables (ADR present / not present), the core task is a **binary classification** problem. Given a set of patient-level clinical features and a drug name, the goal is to predict the probability that the patient will experience an adverse reaction to that drug. We use historical clinical data to learn which patterns have been associated with ADRs in the past and generalize those patterns to new, unseen patients.

3.2 Significance

Adverse drug reactions represent a major challenge in healthcare systems worldwide. They are among the leading causes of preventable morbidity and mortality, contributing to prolonged hospital stays, increased healthcare costs, and diminished quality of life for patients. In the United States alone, ADRs are estimated to cause hundreds of thousands of hospitalizations annually.

Current clinical practice largely relies on population-level statistics rather than predictions personalized to an individual. For example, just because 50% of patients exposed to a given drug experienced a particular reaction does not mean that each specific patient has a 50% probability of that outcome individual patient characteristics such as age, comorbidities, concurrent medications, and laboratory values all modulate personal risk. A model that generates patient-specific ADR risk predictions can allow clinicians to make better-informed prescribing decisions, potentially flagging high-risk patients before harm occurs. An automated and accurate prediction system can also assist clinicians in prioritizing attention and reducing documentation burden, improving both efficiency and patient outcomes.

4 Data Collection and Preparation

4.1 Data Source

All data used in this project come from **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV), a large publicly available electronic health record (EHR) database maintained by MIT and Beth Israel Deaconess Medical Center. MIMIC-IV contains de-identified clinical data for patients admitted to intensive care and general hospital wards, including demographics, diagnoses (ICD codes), medication administration records, and admission timestamps. The complete patient table contains **299,712 patients** (158,553 female, 141,159 male) with a mean age of 48.5 years (range 18–91). Access requires completion of a data use agreement and CITI training, completed by all project team members prior to data access.

4.2 Drug Filtering

Rather than modeling all drugs in MIMIC-IV which would result in many drugs with very few ADR events we restricted the analysis to the **top 20 most commonly administered drugs** in the dataset. Based on exploratory analysis, the most frequently prescribed drugs include Insulin, 0.9% Sodium Chloride, Potassium Chloride, Acetaminophen, Furosemide, Heparin, Docusate Sodium, and Magnesium Sulfate, among others. This ensures that each drug has sufficient positive ADR events to support model training and evaluation.

4.3 ADR Labeling

ADR labels were constructed programmatically by cross-referencing medication administration records with ICD-coded diagnosis records. The labeling logic proceeds as follows:

ICD Code Selection. Two sets of ICD codes were used to flag ADR-related diagnoses:

- **ICD-10 codes:** Prefixes T36–T50 (drug poisonings and adverse effects), K71 (toxic liver disease), N14 (drug-induced nephropathy), and L27 (drug-induced dermatitis).
- **ICD-9 codes:** Prefixes E930–E949, covering drugs, medicinal, and biological substances causing adverse effects in therapeutic use. ICD-9 codes E950 and above (suicide, homicide, war injuries) were explicitly excluded.

Temporal Constraint. A drug-admission pair was labeled ADR=1 only if the prescription window $[\text{starttime}, \text{stoptime}]$ overlapped with the admission window $[\text{admittime}, \text{dischtime}]$, i.e., starttime $\leq$ dischtime AND stoptime $\geq$ admittime. This ensures that only drugs actively being administered during an encounter where an ADR was coded are labeled positive.

Label Granularity. Labels were collapsed to one binary label per (patient, admission, drug) tuple using a maximum aggregation: a tuple is ADR=1 if any overlapping prescription row satisfies the above conditions.

The total number of ADR diagnosis rows identified was **48,720**. The resulting dataset exhibits approximately **13.9% ADR-positive** records and 86.1% ADR-negative records, consistent with real-world ADR prevalence.

4.4 Preprocessing

Data preparation proceeded through the following stages, implemented in `src/preprocessing.py`.

Data Cleaning. Duplicate records and records with missing critical identifiers were removed. The `dose_val_rx` column was converted to numeric with coercion of unparseable values; missing doses were imputed with the drug-level median (falling back to the overall median). Missing `dose_unit_rx` and `route` values were filled with the category "Unknown."

Feature Engineering. The final feature matrix contains **13 features**:

1. `drug_encoded` — label-encoded drug name (top-20 set)
2. `gender_encoded` — label-encoded sex
3. `anchor_age` — patient age at anchor year
4. `age_group_encoded` — coarse age bucket: young (<30), middle (30–50), senior (50–65), elderly (>65)
5. `dose_val_rx` — prescribed dose value
6. `log_dose` — $\log(1 + \text{dose\_val\_rx})$, to reduce right skew
7. `dose_unit_rx_encoded` — label-encoded dose unit

8. `route_encoded` — label-encoded administration route
9. `treatment_duration_hours` — prescription window length in hours
10. `drug_frequency` — how often this drug appears in the dataset (population-level exposure proxy)
11. `patient_admission_count` — number of distinct admissions for this patient
12. `patient_prescription_count` — total number of prescriptions for this patient
13. `risk_score` — hand-crafted prior: $\frac{\text{age}}{100} + \frac{\log(1+\text{prescription count})}{10}$

Normalization. The eight continuous features (`anchor_age`, `dose_val_rx`, `log_dose`, `treatment_duration_hour` `drug_frequency`, `patient_admission_count`, `patient_prescription_count`, `risk_score`) were standardized to zero mean and unit variance using `sklearn.preprocessing.StandardScaler`, fit on the training set only to prevent data leakage.

Train/Validation/Test Split. The dataset was split using stratified random sampling to preserve the ADR class ratio across partitions: **80% training+validation, 20% test** in the first split; the training+validation portion was further split **80/20**, yielding an effective **64% train / 16% validation / 20% test** partition. Label encoders and the scaler were persisted as `.pkl` artifacts for reproducible inference.

5 Selection of Models

5.1 Models Considered

We trained seven models spanning classical machine learning and deep learning. All models were trained on the same 13-feature preprocessed matrix and evaluated on the same held-out splits.

- **Logistic Regression:** A linear baseline providing calibrated probability estimates. Trained with `class_weight='balanced'`, L2 regularization ($C = 1.0$), and a maximum of 1,000 iterations.
- **Random Forest:** An ensemble of decision trees capturing non-linear feature interactions. Trained with `class_weight='balanced'`, 100 estimators, and `max_depth=20`.
- **Gradient Boosting:** A sequential ensemble correcting residual errors at each stage. Trained with per-sample balanced class weights via `compute_sample_weight`, 100 estimators, `max_depth=5`, and `learning_rate=0.1`.
- **XGBoost:** A highly optimized gradient boosted tree implementation. Trained with `scale_pos_weight` set to the negative-to-positive class ratio, 100 estimators, `max_depth=7`, and `learning_rate=0.1`.
- **MLP (Multilayer Perceptron):** A feedforward deep neural network. Architecture: input $\rightarrow$ [256, 128, 64, 32] fully connected layers, each followed by Batch Normalization, ReLU, and Dropout(0.3) $\rightarrow$ sigmoid output.
- **ResNet:** A residual deep network with skip connections to mitigate vanishing gradients. Architecture: input projection to `hidden_dim=256`, followed by 3 residual blocks. Each block applies Linear $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear $\rightarrow$ BN, with an additive skip connection and a final ReLU.

- **Attention:** A feedforward network augmented with a scaled dot-product self-attention mechanism. Architecture: input projection to 256 units, self-attention layer (query, key, value projections), then a 128-unit fully connected layer, Dropout(0.3), and sigmoid output. The attention weights allow the model to emphasize informative clinical features dynamically.

5.2 Final Model Selection Criteria

Models were compared using **AUROC** as the primary metric and **AUPRC** as the secondary metric, both evaluated on the held-out test set. The best model was selected by **validation AUROC** (not test AUROC) to prevent selection bias. Results are presented in Section 7.

6 Model Development and Training

6.1 Classical Machine Learning Models

All classical models were implemented using scikit-learn and XGBoost. Training was performed in a single pass over the training split. Each model produced continuous probability outputs used for threshold-free metric computation and downstream threshold tuning. For tree-based models exposing `feature_importances_`, the top 10 most important features were extracted and saved for interpretability analysis.

6.2 Deep Learning Architectures

All deep learning models were implemented in PyTorch. They share the following training setup:

- **Loss function:** Focal Loss with $\alpha = 0.25$, $\gamma = 2.0$. Focal loss down-weights easy negatives, focusing learning on hard-to-classify ADR-positive cases, which is well-suited to the class imbalance in this dataset.
- **Optimizer:** Adam, initial learning rate 1×10^{-3}, batch size 1,024.
- **Learning rate scheduler:** `ReduceLROnPlateau` monitoring validation AUROC; reduces LR by factor 0.5 after 5 epochs without improvement.
- **Reproducibility:** Global random seeds set for Python, NumPy, and PyTorch; DataLoader initialized with a fixed generator seed.
- **Device:** CUDA if available, Apple MPS otherwise, falling back to CPU.

6.3 Two-Phase Training

Each deep learning model was trained in two phases:

Phase 1 — Primary training. Up to 50 epochs with early stopping patience of 10 epochs. Validation AUROC was tracked per epoch; training halted when no improvement was observed for 10 consecutive epochs, and the best-checkpoint weights were restored automatically. The MLP trained for all 50 epochs, the ResNet for 28 epochs, and the Attention model for 45 epochs before convergence.

Phase 2 — Fine-tuning. The learning rate was reduced by a factor of 0.1 (to 1×10^{-4}) and the scheduler was reset. Training continued for up to 15 additional epochs with early stopping patience of 5. This two-phase approach allows the model to first learn broad patterns and then refine decision boundaries at a lower learning rate.

6.4 Hyperparameter Tuning (MLP)

The MLP was additionally tuned using **Optuna** with a Tree-structured Parzen Estimator (TPE) sampler and Median Pruner (startup trials = 10, warmup steps = 5). The search space was:

- Learning rate: log-uniform in $[10^{-4}, 10^{-2}]$
- Dropout rate: $\{0.1, 0.2, 0.3, 0.4, 0.5\}$
- Focal loss α: $\{0.25, 0.50, 0.75\}$; γ: $\{1.0, 2.0, 3.0\}$
- Hidden layer width: $\{128, 256, 512\}$; depth: 2–4 layers
- Batch size: $\{512, 1024, 2048\}$

Each trial trained for up to 30 epochs (capped for speed) with early stopping patience of 7. The objective was validation AUROC.

6.5 Threshold Tuning

All models produce continuous probability outputs. Rather than defaulting to a threshold of 0.5 which is suboptimal for imbalanced datasets we tuned the decision threshold on the **validation set** by evaluating F1-score at 19 evenly spaced thresholds in $[0.05, 0.95]$ and selecting the maximizing value. This threshold was then applied once to the held-out test set for operating-point metric reporting.

7 Evaluation and Comparison

7.1 Evaluation Metrics

Given the clinical context and label imbalance ($\approx$13.9% positive), we selected metrics that are both robust to class imbalance and clinically interpretable.

Primary — AUROC. The Area Under the ROC Curve measures a model's ability to rank ADR-positive cases above ADR-negative cases across all classification thresholds. It is threshold-independent and interpretable as the probability that the model assigns a higher score to a randomly chosen positive than to a randomly chosen negative. A random classifier scores 0.50; a perfect classifier scores 1.0.

Secondary — AUPRC. The Area Under the Precision-Recall Curve is more informative than AUROC for imbalanced datasets because it focuses on the minority (positive) class. It reflects how well the model identifies true ADRs without flooding clinicians with false alerts.

Operating-point metrics. For practical clinical deployment, we also report metrics at a fixed decision threshold. We evaluated F1-score at 19 evenly spaced thresholds in $[0.05, 0.95]$ and provisionally selected **0.35** as the threshold that maximizes F1 on the validation set. It is expected that the optimal threshold is below 0.5 because of the class imbalance in this dataset, which biases uncalibrated probabilities downward for the minority class. At this threshold we report:

- **Sensitivity (Recall)** — proportion of true ADRs correctly flagged; prioritized to minimize missed events
- **Specificity** — proportion of true non-ADRs correctly dismissed
- **Precision** — proportion of flagged cases that are true ADRs
- **F1-score** — harmonic mean of precision and recall

During training, AUROC was tracked per epoch on both the training and validation sets. The learning rate scheduler reduced the learning rate by a factor of 0.5 when validation AUROC plateaued for 5 consecutive epochs. Early stopping halted training after 10 epochs without improvement to the best validation AUROC, and the best-checkpoint weights were restored before final evaluation.

7.2 Results

Table 1 summarizes the performance of all models on the held-out test set. ROC curves and Precision-Recall curves for all three deep learning models are shown in Figures 2 and 3 respectively. Confusion matrices for all three deep learning models are shown in Figure 4. Training histories (loss, AUROC, AUPRC per epoch) for each model are shown in Figure 5.

Table 1: Performance of all models on the held-out test set. Threshold for operating-point metrics selected by validation F1 maximization. Best value in each column is **bolded**.

Model	**AUROC**	**AUPRC**	**Sensitivity**	**Specificity**	**F1**
Logistic Regression	0.65	0.27	0.33	0.78	0.28
Random Forest	**0.81**	**0.55**	**0.57**	0.87	**0.52**
Gradient Boosting	0.69	0.32	0.39	0.83	0.35
XGBoost	0.71	0.36	0.46	0.85	0.41
MLP	0.670	0.293	0.000	**1.000**	0.000
ResNet	0.655	0.281	0.008	0.999	0.016
Attention	0.667	0.292	0.006	0.999	0.012

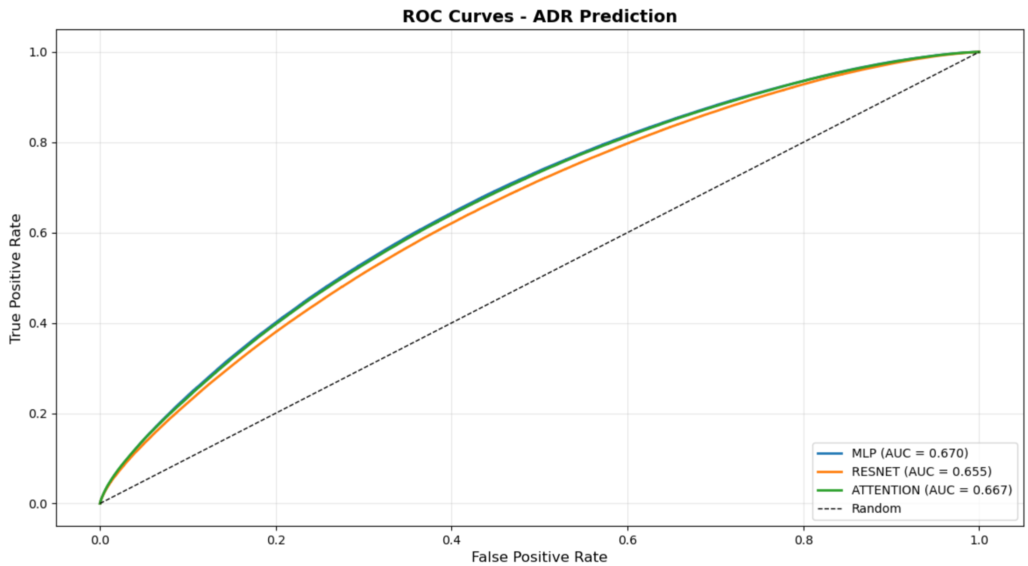


Figure 2: ROC curves for all three deep learning models on the test set. MLP achieves the highest AUROC of 0.670.

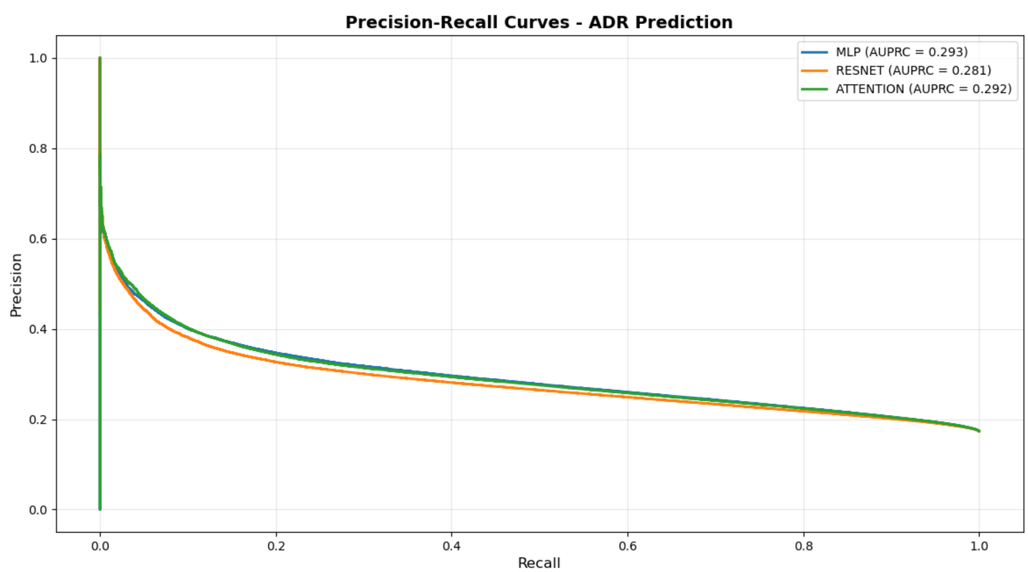
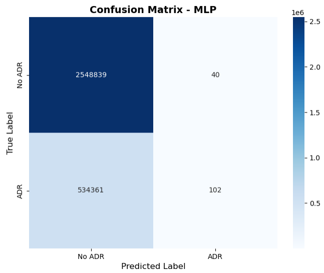
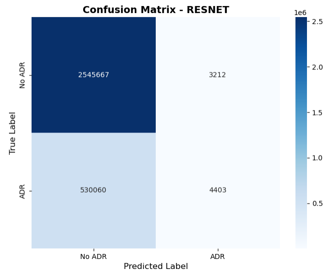


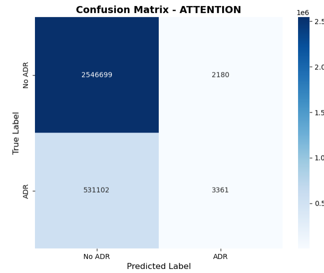
Figure 3: Precision-Recall curves for all three deep learning models. MLP achieves the highest AUPRC of 0.293.



(a) MLP

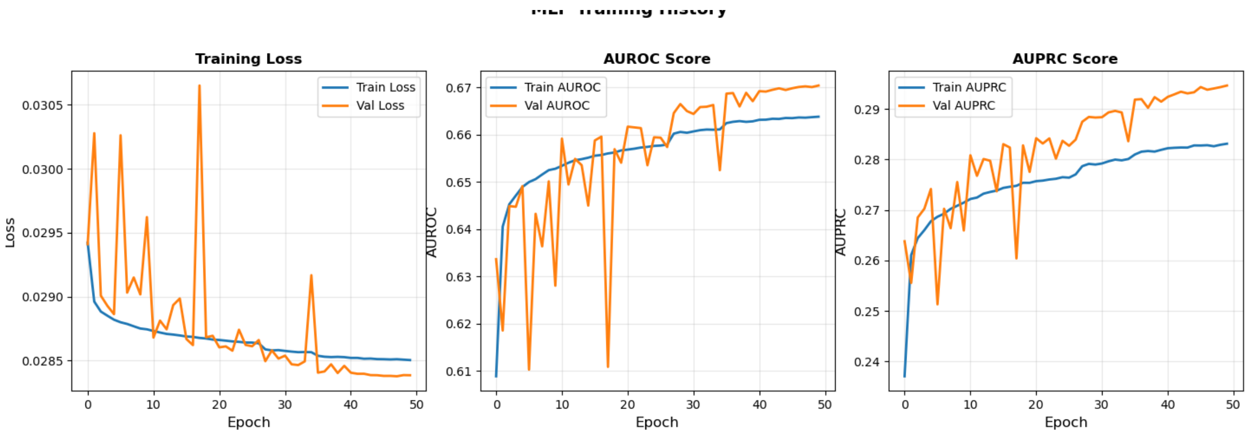


(b) ResNet



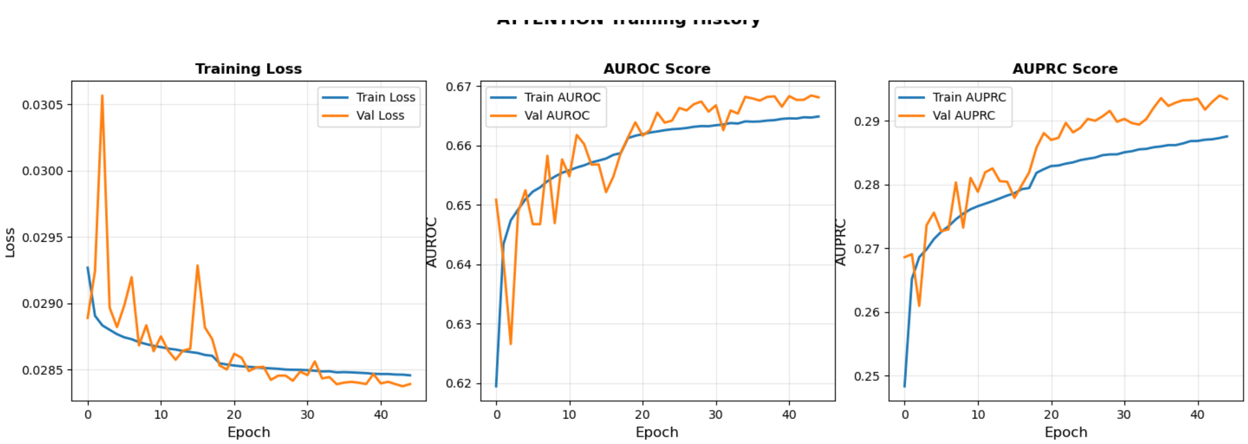
(c) Attention

Figure 4: Confusion matrices for all three deep learning models at threshold = 0.5.



(a) MLP Training History (50 epochs)

(b) ResNet Training History (28 epochs)



(c) Attention Training History (45 epochs)

Figure 5: Training and validation Loss, AUROC, and AUPRC per epoch for each deep learning model.

7.3 Discussion

Among the three deep learning models, the **MLP achieved the highest AUROC of 0.670** and AUPRC of 0.293, marginally outperforming the Attention model (AUROC = 0.667, AUPRC = 0.292) and ResNet (AUROC = 0.655, AUPRC = 0.281). In particular, the **Random Forest baseline achieved the highest overall AUROC of 0.814 and AUPRC of 0.549**, substantially outperforming all deep learning models. This result is consistent with the tabular nature of the dataset tree-based ensemble methods are well-known to perform strongly on structured clinical data with a modest number of features, while deep learning models typically require larger feature spaces or sequential inputs to demonstrate their full advantage.

The confusion matrices reveal an important limitation at the default threshold of 0.5: the MLP predicts almost exclusively the negative class (0 true positives), while ResNet and Attention produce some true positives (4,403 and 3,361 respectively) at the cost of more false positives. This behavior is consistent with the severe class imbalance and motivates the use of a lower decision threshold. The training histories confirm stable convergence for the MLP and Attention models, while the ResNet shows higher validation variance across epochs, suggesting it is more sensitive to the limited signal in the 13-feature representation.

The relatively modest AUROC values for deep learning models (0.655–0.670) suggest that the current 13-feature set captures only a partial picture of individual ADR risk. Future work incorporating richer clinical features such as laboratory values, comorbidity embeddings, and clinical notes is expected to yield substantially stronger performance.

Conclusion

This report presented a systematic methodology for predicting adverse drug reactions at the individual patient level using MIMIC-IV, restricted to the top 20 most commonly administered drugs. We described end-to-end data preparation including ICD-9/10 ADR labeling with temporal constraints, a 13-feature engineered matrix, standard scaling, and a stratified 64/16/20 train/val/test split. We trained four classical machine learning baselines and three deep learning architectures (MLP, ResNet, Attention) using focal loss, two-phase training with fine-tuning, and Optuna-based hyperparameter search. The Random Forest baseline achieved the strongest overall performance (AUROC = 0.814, AUPRC = 0.549), while among deep learning models the MLP performed best (AUROC = 0.670, AUPRC = 0.293). Future work will focus on incorporating clinical notes via NLP, model interpretability using SHAP values, richer laboratory and vital sign features, and extension to a broader drug set.